

P. E. S. COLLEGE OF ENGINEERING

Mandya, Karnataka – 571401

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DEPARTMENT HANDBOOK

Department of Computer Science & Engineering

Academic Year 2025 – 2026

Affiliated to VTU, Belagavi | Autonomous – UGC | NAAC A Grade | NBA Accredited
B.E. (CSE) | B.E. (AI & ML) | B.E. (Data Science) | B.E. (CSBS) | Ph.D.

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Chapter 1: Department Overview

1.1 About the Department

Established in 1983, the Department of Computer Science and Engineering (CS&E) at PES College of Engineering, Mandya is committed to delivering quality education and shaping future-ready professionals. With a strong emphasis on both theoretical foundations and practical skills, the department equips students to thrive in a rapidly evolving tech landscape.

The programme follows an Outcome-Based Education (OBE) model under a Choice-Based Credit System (CBCS), allowing students to personalise their learning journey and pursue their career goals with confidence. With an annual intake of 240 students in the core CSE programme alone, the department is powered by a team of highly qualified and experienced faculty members with expertise across various domains of Computer Science. The department also offers a Ph.D. programme in Computer Science and Engineering, introduced in 2004, fostering advanced research and innovation.

1.2 Department at a Glance

Established	1983
Parent Institution	P. E. S. College of Engineering, Mandya (Est. 1962)
Affiliation	Visvesvaraya Technological University (VTU), Belagavi
Autonomous Status	UGC Autonomous (2008-09)
Accreditation	NAAC – A Grade NBA Accredited
UG Intake (CSE)	240 students per year
UG Intake (AI & ML)	60 students per year
UG Intake (Data Science)	60 students per year
UG Intake (CSBS)	60 students per year
PG Programme	M.Sc. (Engineering) by Research
Research Programme	Ph.D. (introduced 2004)
Programme Duration	4 Years (UG) 2 Years (PG)
Programme Type	Outcome-Based Education (OBE) + CBCS
Ph.D Guides	6 recognised research guides
Ph.D Awarded	12+ from the department
Active Researchers	30+ scholars
Website	https://pesce.ac.in/departments-computer-science.php

1.3 Highlights

- Industry-aligned, regularly updated curriculum.
- Experienced, research-oriented, and industry-connected faculty.
- Hands-on labs and project-based learning environment.
- Strong industry connect and placement support.
- Active professional clubs: IEEE, ISTE, CSI, GLUG, Hack Club.
- IoT Centre of Excellence and Data Science Lab on campus.
- Multiple funded research projects and published patents.
- MoU with global partners including MongoDB and Intel.

Chapter 2: Vision, Mission & Programme Outcomes

2.1 Vision

"The Department of Computer Science and Engineering shall create professionally competent and socially responsible engineers capable of working in a global environment."

2.2 Mission

Code	Mission Statement
DM1	Enforce best practices in teaching-learning, with dedicated faculty and supportive infrastructure to impart the knowledge in emerging technologies — required to create professionally competent engineers.
DM2	Improve Industry-Institute relationship for mutual benefit — required to create professionally competent engineers.
DM3	Inculcate ethical values, communication, and entrepreneurial skills — required to create professionally competent and socially responsible engineers.
DM4	Sensitise social, legal, environmental, and cultural diversity issues through professional training and balanced curriculum — required to create engineers capable of working in a global environment.

2.3 Programme Educational Objectives (PEOs)

Graduates of the programme will be able to:

PEO	Objective
PEO 1	Have a successful computer professional career in IT industry and related areas.
PEO 2	Pursue higher education in engineering or management with a focus on intensive research and developmental activities.
PEO 3	Develop their career as entrepreneurs in a responsible, professional, and ethical manner to serve the society.

2.4 Programme Outcomes (POs)

Engineering graduates of the CS&E programme will be able to:

PO	Description
PO1 – Engineering Knowledge	Apply knowledge of mathematics, natural science, computing, and engineering fundamentals to develop solutions to complex engineering problems.

PO2 – Problem Analysis	Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions with consideration for sustainable development.
PO3 – Design/Development of Solutions	Design creative solutions for complex engineering problems considering public health, safety, culture, society, and environment.
PO4 – Investigations	Conduct investigations of complex engineering problems using research-based knowledge including experimental design, modelling, and data interpretation.
PO5 – Engineering Tool Usage	Create, select, and apply appropriate techniques, resources, and modern IT tools to solve complex engineering problems.
PO6 – Engineer and World	Analyse and evaluate societal and environmental aspects of complex engineering solutions with reference to sustainability, economy, health, and legal frameworks.
PO7 – Ethics	Apply ethical principles and commit to professional ethics, human values, diversity, and inclusion; adhere to national and international laws.
PO8 – Teamwork	Function effectively as an individual and as a member or leader in diverse/multi-disciplinary teams.
PO9 – Communication	Communicate effectively within the engineering community and society, comprehend and write effective reports, and make effective presentations.
PO10 – Project Management & Finance	Apply engineering management principles and economic decision-making to manage projects in multidisciplinary environments.
PO11 – Life-Long Learning	Recognise the need for independent and life-long learning, adaptability to new technologies, and critical thinking in the context of technological change.

2.5 Programme Specific Outcomes (PSOs)

PSO	Outcome
PSO-1	Ability to apply problem-solving skills in developing solutions through fundamentals of Computer Science and Engineering.
PSO-2	Ability to apply analytical skills in the field of Data Processing Systems.
PSO-3	Ability to design and develop applications through Software Engineering methodologies and Networking Principles.

2.6 Knowledge and Attitude Profile (WK Framework)

Code	Knowledge Area
WK1	Systematic, theory-based understanding of natural sciences applicable to the discipline and awareness of relevant social sciences.
WK2	Conceptually-based mathematics, numerical analysis, data analysis, statistics, and formal aspects of computer and information science.
WK3	Systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.

WK4	Engineering specialist knowledge providing theoretical frameworks and bodies of knowledge for accepted practice areas.
WK5	Knowledge of efficient resource use, environmental impacts, whole-life cost, re-use of resources, net-zero carbon, and similar sustainability concepts.
WK6	Knowledge of engineering practice (technology) in practice areas of the engineering discipline.
WK7	Knowledge of the role of engineering in society and identified issues such as professional responsibility and sustainable development.
WK8	Engagement with current research literature; awareness of critical thinking and creative approaches to evaluate emerging issues.
WK9	Ethics, inclusive behaviour, and conduct. Knowledge of professional ethics, responsibilities, norms, and awareness of the need for diversity.

Chapter 3: Academic Programmes & Syllabus

3.1 Programmes Offered

Programme	Degree	Intake	Duration
Computer Science & Engineering (CSE)	B.E.	240	4 Years
Computer Science & Engineering (AI & ML)	B.E.	60	4 Years
Computer Science & Engineering (Data Science)	B.E.	60	4 Years
Computer Science & Business System (CSBS)	B.E.	60	4 Years
M.Sc. (Engineering) by Research	PG	–	2 Years
Doctor of Philosophy (Ph.D.)	Ph.D.	–	3–5 Years

3.2 Lateral Entry

Students holding a 3-year Diploma in Computer Science or related branch are eligible for lateral entry into the 2nd year of the B.E. (CSE) programme at 10% of approved intake.

3.3 Syllabus Schemes

Syllabus Document	Scheme	Available At
I Year Syllabus	2021 Scheme	https://pesce.ac.in/admin/photo/3/syllabuspdf/19.pdf
I Year Syllabus	2022 Scheme	https://pesce.ac.in/admin/photo/3/syllabuspdf/105.pdf
I Year Syllabus	2025 Scheme	https://pesce.ac.in/admin/photo/3/syllabuspdf/169.pdf
II Year Syllabus	2021 Scheme	https://pesce.ac.in/admin/photo/3/syllabuspdf/21.pdf
II Year Syllabus	2022 Scheme	https://pesce.ac.in/admin/photo/3/syllabuspdf/22.pdf
II Year Syllabus	2024 Scheme	https://pesce.ac.in/admin/photo/3/syllabuspdf/170.pdf
III Year Syllabus	2021 Scheme	https://pesce.ac.in/admin/photo/3/syllabuspdf/25.pdf
III Year Syllabus	2022 Scheme	https://pesce.ac.in/admin/photo/3/syllabuspdf/104.pdf
IV Year Syllabus	2022 Scheme	https://pesce.ac.in/admin/photo/3/syllabuspdf/166.pdf

3.4 Programme Duration & Batch Cycles

Programme Name	Duration	Batch Start	Batch End
Computer Science & Engineering	3 years	2021	2024
Computer Science & Engineering	3 years	2024	2027

Chapter 4: Faculty Profiles

The department is powered by 30+ teaching faculty members with expertise spanning Natural Language Processing, Machine Learning, IoT, VLSI, Image Processing, Cloud Computing, Software Engineering, and more. Detailed profiles of each faculty member are presented below.

4.1 Professors

Dr. Anitha M L Professor & Head of Department (HOD)	
Designation	Professor & HOD, Dept. of Computer Science & Engineering
Email	anithaml@pesce.ac.in Contact: +91 9945576186
Qualifications	Ph.D – University of Mysore (PESCE, Mandya) M.Sc. (Engg.) by Research – VTU, Belagavi (PESCE) B.E. – University of Mysore (PESCE)
Research Areas	Biometric Systems, Image Processing, IoT-based Automation, Deep Learning for Medical Imaging, Hydroponic Systems
Notable Publications	IoT-Based Automated Hydroponic System (ICERECT 2022); Segmentation of Retinal Images using MesU-Net (IJOBE 2023); Disease Grading of Diabetic Retinopathy (ICCMC 2022); Biometric Verification using Finger Knuckle Print & Hand Geometry (WCE 2016)
Patents Filed	Predicting Child Mental Stress using AI – App No: 202341016400 (Published)
IEEE Role	COMSOC Mentor, IEEE-PESCE Student Branch

Dr. M C Padma Professor	
Designation	Professor, Dept. of Computer Science & Engineering
Email	padmamc@pesce.ac.in Contact: +91 9141113030
Qualifications	Ph.D – Visvesvaraya Technological University (VTU), Belagavi M.Sc. Tech – University of Mysore B.E. – Computer Science & Engineering, PESCE, Mandya
Research Areas	Natural Language Processing (NLP), Data Mining & Analytics, Artificial Intelligence & Machine Learning, Sentiment Analysis, Machine Translation
Ph.D Scholars Guided	10+ scholars awarded Ph.D (VTU and University of Mysore); 10+ scholars ongoing
CSI Role	Founded CSI-Student Chapter at PESCE (2015); Former HOD of CSE Department
IEEE Role	WIE Society Mentor, IEEE-PESCE Student Branch
Notable Publications	OAHRSAM: Optimized Automatic High Rank Sentiment Analysis Model (Elsevier 2022); Insights of Search Engine Optimization using NLP & ML (IJACSA 2023); Automatic Sentiment Examination Model (Turkish Journal of Computer & Mathematics Education 2021)

Dr. Vinay S Professor & Vice Principal / Placement Officer	
Designation	Professor, Vice Principal & Placement Officer

Email	vinay@pesce.ac.in Contact: +91 9986515835
Qualifications	Ph.D – Decision Support System in Requirements Engineering, Manipal University M.Tech – Computer Engineering, SJCE, Mysuru B.E. – Computer Science & Engineering, PESCE, Mandya
Research Areas	Machine Learning, Software Engineering, Sentiment Analysis in Kannada, Cloud Computing, Anomaly Detection
Scopus ID	56604895600
ORCID	https://orcid.org/0000-0002-1698-9314
Funded Project	New Age Incubation Network (NAIN) – ₹1.2 Crore, Government of Karnataka (2021–2023)
Patents Filed	Disease Detection in Chrysanthemum using Deep Learning (App No: 202441029970 A – Published); AI-enabled Solar Panel Rotation System (App No: 202341016403 – Published); Predicting Child Mental Stress using AI (App No: 202341016400 – Published)
IEEE Role	Branch Mentor, IEEE-PESCE Student Branch
Notable Publications	Enhancing Sentiment Analysis in Kannada Texts by Feature Selection (IJECE Vol 14, 2024); Caste and Migration Hate Speech Detection (LT-EDI 2024); A Novel Analytical Framework for Educational Intelligence-as-a-Service (Wireless Personal Communications, Springer, 2022)

Dr. H P Mohan Kumar Professor	
Designation	Professor, Dept. of Computer Science & Engineering
Email	mohankumar@pesce.ac.in
Qualifications	Ph.D – Computer Science, University of Mysore (2015) M.Sc. (Tech) – PET Research Centre, University of Mysore (2009) MCA – PESCE, University of Mysore (1998)
Research Areas	Computer Vision, Machine Learning, Gait Recognition, Deep Learning, Image Processing
Research Guide	Recognised Ph.D guide at VTU
Notable Publications	Mobile Based Classification of Diabetic Retinopathy Using TinyML (IEEE NKCon 2023); Gait Recognition Based on Symbolic Representation (Int. Journal of Machine Intelligence 2011); Vision for Blind (IJARCCE 2024); 35+ journal and conference publications

Dr. Nagarathna Professor	
Designation	Professor, Dept. of Computer Science & Engineering
Email	nagarathna@pesce.ac.in
Qualifications	Ph.D – Sensor Networks, Anna University, Chennai M.Tech – Computer Science & Engineering, VTU, Belagavi
Research Areas	Sensor Networks, IoT, Data Science, Machine Learning, Agricultural AI, Deep Learning
Ph.D Scholars Guided	3+ scholars awarded Ph.D (VTU)

Patents Filed	Automated System to Predict Air Quality Index in Layer Poultry Farm Using Deep Learning (App No: 202441027160 A – Published); AI Enabled Solar Panel Rotation System (App No: 202341016403 – Published)
Notable Publications	YOLO-Based Deep Learning Models for Insect Detection (Agriculture, MDPI Vol 13, 2023); Comparative Performance Evaluation of Classification Models for Flying Insects (INDECS Vol 21, 2023); Attention-Based Convolutional dual sparse autoencoder for Poultry Farm Analysis (Multimedia Tools & Applications, Springer, 2024)

4.2 Associate Professors

Prasanna P Associate Professor	
Designation	Associate Professor, Dept. of Computer Science & Engineering
Email	shivz3@yahoo.com
Qualifications	M.Tech – SJCE, Mysore
Research Areas	Agri-robotics, Smart ATM Security, Intelligent Traffic Systems
Patents Filed	Olympics Game Analysis Tool Using Machine Learning (App No: 202441027159 A – Published); Modeling and Formal Specification of Smart Mass Transit Railway Interlocking System (App No: 202441027225 A – Published)
Notable Publications	Automated Agribusiness Furrowing Seeding and Grass Cutting (IJRCCE 2017); Smart Traffic Management for Emergency Vehicle (IJCSMC 2018)

Dr. Veena M Associate Professor	
Designation	Associate Professor, Dept. of Computer Science & Engineering
Email	mveenakemps@gmail.com
Qualifications	Ph.D – VTU, Belagavi M.Tech – Software Engineering, SJCE, Mysuru B.E. – Computer Science & Engineering, PESCE, Mandya
Research Areas	Image Processing, Medical Imaging, Breast Cancer Detection, IoT in Healthcare
Ph.D Status	Awarded Ph.D from VTU (2023) under Dr. M.C. Padma – Thesis: Techniques for Early Detection of Breast Cancer using Multimodal Images
Notable Publications	Detection of Region of Interest and Stages using Mammogram Images (IJARCCE 2022); Fusion of Features from Mammogram and DBT Views (ICCR 2021); Detecting Stages of Breast Cancer using CNN (Vol 11, Issue 5, 2024)

Dr. Prasanna Kumar R S Associate Professor	
Designation	Associate Professor, Dept. of Computer Science & Engineering
Email	prasannapesce@yahoo.co.in
Qualifications	Ph.D – Choudhary Charan Singh University

Research Areas	IoT, Machine Learning, Big Data, Wireless Sensor Networks (WSN), Disaster Prediction
Notable Publications	Landslide, Flood Prediction Using WSN Data (IJCRT, March 2020); Earthquake Prediction Using WSN Data and Machine Learning (IJCSE, March 2020)

Dr. Girish Babu MC | Associate Professor

Designation	Associate Professor, Dept. of Computer Science & Engineering
Email	mcgirishbabu@gmail.com
Qualifications	Ph.D – VTU, Belagavi M.Tech – NIE, Mysore, VTU B.E. – PESCE, Mandya, VTU
Research Areas	Image Processing, Hyperspectral Crop Classification, Remote Sensing, Deep Learning
Ph.D Status	Awarded Ph.D from VTU (2024) under Dr. M.C. Padma – Thesis: Techniques for Identification & Classification of Crops using Satellite Imagery
Notable Publications	Semantic Feature Extraction for Hyperspectral Crop Classification (Indonesian Journal of Electrical Engineering & CS 2021); Inherent Feature Extraction using Soft Margin Decision Boundary for Crop Classification (IJACSA 2021)

Dr. H R Divakar | Associate Professor

Designation	Associate Professor, Dept. of Computer Science & Engineering
Email	divakarhr@pesce.ac.in
Qualifications	Ph.D – Sri Siddhartha Academy of Higher Education, Tumakuru
Research Areas	Data Mining, Machine Learning, Deep Learning, Healthcare Predictive Analytics
Notable Publications	Prediction of Cervical Cancer with Ontology Based Deep Learning (IJCSC 2020); An Ontology-Based System to Predict Diabetes Using Deep Learning (Dogo Rangsang Research Journal 2020); Machine Learning Approaches to Predict Diabetes Mellitus – A Survey (AEGAEUM Journal 2020)

Dr. Deepika | Associate Professor

Designation	Associate Professor, Dept. of Computer Science & Engineering
Email	deepika@pesce.ac.in
Qualifications	Ph.D – VTU (under Dr. Nagarathna) – Thesis: Predictive Model for Production Enhancement in Layer Poultry Farm using Machine Learning Techniques (Awarded Feb 2025)
Research Areas	Deep Learning, IoT, Poultry Farm Analytics, Air Quality Prediction, Medicinal Plant Recognition
Patents Filed	Automated System to Predict Air Quality Index in Layer Poultry Farm (App No: 202441027160 A – Published); Mobile App for Recognizing Ayurvedic Medicinal Plants using Deep Learning (App No: 202441027221 A – Published)
Funded Project	IoT Project – ₹1,30,000, Government of Karnataka (2024)

Notable Publications

Ensemble Tree Classifier based Analysis of Water Quality (IJACSA Vol 13, 2022);
Leaf Blast Rice Disease Prediction Using Binary Classifiers (IJCSIT 2021)

4.3 Assistant Professors

The following assistant professors contribute to teaching, research guidance, and departmental activities:

#	Name	Designation	Research / Ph.D Status
1	S K Shivashankar	Assistant Professor	Pursuing Ph.D at VTU (Guide: Dr. Mahesh Kaluti) – MANET Routing using ANN & ML
2	Chaitra B P	Assistant Professor	Pursuing Ph.D at VTU (Guide: Dr. M L Anitha)
3	Shwetha M K	Assistant Professor	Pursuing Ph.D at VTU (Guide: Dr. Nagarathna) – Rice Disease Prediction
4	Bhavya D	Assistant Professor	Course work completed, JSS Mysore (Guide: Dr. D S Vinod) – Network Security & Heterogeneous Networks Patent filed: Root Cause Analysis Device for Heterogeneous Networks (App No: 202141000707)
5	C Chethana	Assistant Professor	Pursuing Ph.D at VTU (Guide: Dr. Deepu Singh) Patent: Automated Driving Assistance System (App No: 202141019449)
6	Suraj B S	Assistant Professor	Ph.D at VTU (Guide: Dr. M C Padma) – Sentiment Analysis for Fake Reviews CV Completed
7	Sindhu B S	Assistant Professor	Registered for Ph.D at VTU (Guide: Dr. Minavathi) – Microscopic Image Analysis for Cancer Diagnosis using Generative AI
8	Raghavendra Babu TM	Assistant Professor	Teaching & Research
9	Pramod Kumar B M	Assistant Professor	Patent co-inventor: AI-Enabled Solar Panel Rotation System (App No: 202341016403 – Published)
10	Poojitha N	Assistant Professor	Teaching & Research
11	B J Srujana	Assistant Professor	Pursuing Ph.D (Guide: Dr. Vinay S) – Disease Detection in Chrysanthemum using DL Patent: Disease Detection in Chrysanthemum (App No: 202441029970 A – Published)
12	Namratha H T	Assistant Professor	Teaching & Research
13	Thejashwini M	Assistant Professor	Teaching & Research
14	Hemalatha B	Assistant Professor	Teaching & Research
15	Sukanya H N	Assistant Professor	Teaching & Research
16	VedaShree YC	Assistant Professor	Teaching & Research
17	Manasa S	Assistant Professor	Teaching & Research

18	Megha A	Assistant Professor	Teaching & Research
19	Kousthubha HR	Assistant Professor	Teaching & Research
20	Rashmi Y S	Assistant Professor	Teaching & Research

4.4 Technical & Non-Teaching Staff

#	Name	Designation
1	M S Sunita	Computer Operator
2	K C Sumalatha	Computer Operator
3	M K Vidya Shree	Computer Instructor
4	ChinneGowda H S	Assistant Instructor
5	K Sreedhar Prasad	System Administrator
6	Vinutha H N	SDA / Computer Operator
7	Preetham Edigar M S	System Administrator
8	Shubha	System Instructor
9	Manushree K M	Computer Operator
10	Pavithra D C	Computer Operator
11	Shwetha K	Computer Operator
12	Ravichandra K M	FDA / Computer Operator
13	Vivekananda M N	Helper
14	Mahesh Kumar H K	Helper / Mechanic
15	Manohara H S	Helper
16	Manjunath H N	Helper
17	Shobha	Helper
18	Sindhuja H M	Helper
19	Prema	Helper
20	Akash S S	Helper

Chapter 5: Research & Innovation

5.1 Research Guides

Name	Designation	Email
Dr. M C Padma	Professor	padmamc@pesce.ac.in
Dr. R Girish	Professor	write2girisha@gmail.com
Dr. Nagarathna	Professor	nagarathna@pesce.ac.in
Dr. H P Mohan Kumar	Professor	mohankumar@pesce.ac.in
Dr. Vinay S	Professor	vinay@pesce.ac.in
Dr. M Veena	Associate Professor	mveenakemps@gmail.com

5.2 Ph.D Degrees Awarded

Scholar	University	Date of Award	Guide	Thesis Title
Prathibha R J	VTU	20/04/2018	Dr. M C Padma	Novel Approaches to Develop Modules of Source Analysis for Kannada Sentence in Machine Translation System
Mohammad Ahmed Ail Abdu Alrhman	UOM	18/06/2019	Dr. M C Padma	Design of Robust Concept Drift Adoption Strategies for High Speed Data Stream Mining
Arjun U	VTU	08/04/2022	Dr. Vinay S	An Approach to Enable the Data Integrity for Cloud Data Storage by Auditing
Manasa K N	UOM	19/05/2023	Dr. M C Padma	A Framework for Analysing the Intensity of Sentiments in Social Media
Veena M	VTU	26/07/2023	Dr. M C Padma	Techniques for Early Detection of Breast Cancer using Multimodal Images
Esraa Mohamand Jamil Alsariera	UOM	30/12/2023	Dr. Padma M C	–
Dr. M C Girish Babu	VTU	22/02/2024	Dr. M C Padma	Techniques for Identification & Classification of Crops using Satellite Imagery
Dr. Chethan G S	VTU	30/04/2024	Dr. Vinay S	Design of Dependable Architecture for Collaborative Platform of Educational Big Data on Cloud Environment
Dr. Chowdary K K	VTU	23/08/2024	Dr. R Girisha	Regional Climate Change Study using HPC and Data Mining

Dr. Nitin Kumar	VTU	23/09/2024	Dr. Nagarathna	Early Prediction and Classification of Mango Fruit Flies with Seasonal Activity using Sensor Network
Deepika	VTU	08/02/2025	Dr. Nagarathna	Predictive Model for Production Enhancement in Layer Poultry Farm using Machine Learning Techniques

5.3 Active Research Scholars

Scholar	University	Guide	Research Topic	Status
Suraj B S	VTU	Dr. Padma M C	Sentiment Analysis Model for Recognizing Fake Reviews using Opinion Mining	CV Completed
Prathibha S	VTU	Dr. Vinay S	ML Approach for Anomaly Detection System in Cloud	CV Completed
Raghavendra K	VTU	Dr. Vinay S	Swarm Intelligence based Approach for Sensor Selection Clustering in IoT	Open Seminar Pending
Sunil M E	VTU	Dr. Vinay S	Analysis of Subjective Information in Public Platforms Using ML	Thesis Submission Pending
Chethan Kumar V	VTU	Dr. D R Umesh	Assistive System for Intellectually Disabled Children using Big Data Analytics over WBAN	Final Viva Pending
Shruthi P S	VTU	Dr. D R Umesh	Multidimensional Analysis in Analytics for Quality Optimization	Final Viva Pending
Shivashankar S K	VTU	Dr. Mahesh Kaluti	Enhancing Routing Protocol Efficiency in MANETs using ANN & ML	Course Work Exam 2022
Meghana M	VTU	Dr. Nagarathna	Predicting Legal Outcome using AI and Deep Learning	Course Work Exam 2022
Raksthitha M S	VTU	Dr. Vinay S	Traceability System based on Blockchain	Pursuing Course Work
Vishesh J	VTU	Dr. R Girisha	Customer Behaviour Analysis using Historical Purchase Data and AI	CV Completed
Dinesh S D	VTU	Dr. Vinay S	Gait Pattern Analysis for Identification using Topic Model	Colloquium Completed
Sachith B K	VTU	Dr. D R Umesh	Curvilinear Text Detection and Recognition from Multi-View Image Scenes	CV Completed
Sindhu P	VTU	Dr. Padma M C	Digital Forensics-Based Malware Detection Using Machine Learning	Course Work Completed
Srujana B J	VTU	Dr. Vinay S	Disease Detection and Classification in Chrysanthemum Flowers using Deep Learning	Course Work Completed

Vinutha M S	UOM	Dr. Padma M C	Search Engine Optimization using NLP and ML	Pursuing
Sindhu B S	VTU	Dr. Minavathi	Microscopic Image Analysis for Cancer Diagnosis using Generative AI	Registered
Bhavya M R	VTU	Dr. R Girish	Computational Model for Brain Tumour Detection using Deep Learning	CV Completed
Varsha N	UOM	Dr. Padma M C	Secure User Engagement in Online Social Networks using Deep Contagion & Graph Mining	Pursuing Course Work
Fahad Abdullah Ghanem Mohammed Nasser	VTU	Dr. Vinay S	Automatic Short Text Summarization for Social Media Data	Thesis Submission Pending
Sharaf H M Almosalami	UOM	Dr. Padma M C	–	Pursuing Course Work

5.4 Faculty Research (Pursuing Ph.D)

Faculty Name	Guide	University	Status
R S Prasanna Kumar	Dr. S N Thimmaraju	VTU	Course Work Completed
Suraj B S	Dr. M C Padma	VTU	Registered – Preparing for Course Work
Deepika	Dr. Nagarathna	VTU	Course Work Completed (Awarded 2025)
Shwetha M K	Dr. Nagarathna	VTU	Registered – Preparing for Course Work
Chaitra B P	Dr. M L Anitha	VTU	Registered – Preparing for Course Work
V Chethan Kumar	Dr. D R Umesh	VTU	Registered – Preparing for Course Work
Shruthi P S	Dr. D R Umesh	VTU	Registered – Preparing for Course Work
C Chethana	Dr. Deepu Singh	VTU	Registered – Preparing for Course Work
Bhavya D	Dr. D S Vinod	JSS, Mysore	Course Work Completed
Jayshankar	–	IIT Varanasi	Course Work Completed
Yoga B S	–	IIT Roorkee	Registered
Ramyashree H P	Dr. Minavathi	VTU	Comprehensive Viva Completed

5.5 Patents Filed & Published

Faculty / Inventors	Patent Title	App. No.	Status
Deepika, Dr. Nagarathna	Automated System to Predict Air Quality Index in Layer Poultry Farm Using Deep Learning	202441027160 A	Published
Ms. Srujana B J, Dr. Vinay S	Disease Detection and Classification in Chrysanthemum Flowers using Deep Learning	202441029970 A	Published
Deepika	Mobile Application Development for Recognizing Ayurvedic Medicinal Plants using Deep Learning	202441027221 A	Published
P Prasanna, Dr. Umesh D R, S K Shivashankar	Olympics Game Analysis Tool Using Machine Learning	202441027159 A	Published
Uma S K	Sign Language Recognition System for Hearing and Speech Impaired People Using ML	202441027157 A	Published
Dr. Umesh D R, P Prasanna, Chetan Kumar V	Modeling and Formal Specification of Smart Mass Transit Railway Interlocking System	202441027225 A	Published
Dr. Nagarathna, Dr. Vinay S, Mr. Pramod Kumar B M	An AI-Enabled Solar Panel Rotation System	202341016403	Published
Dr. Anitha M L, Dr. Vinay S	Predicting Child Mental Stress using AI	202341016400	Published
Dr. D R Umesh	IoT-Enhanced Alternate Pair Furrow Irrigation System for Sustainable Water Management in Sugarcane	202341082020	Published
Bhavya D	Root Cause Analysis, Threat Interpretation, and Network Survivability Prediction Device for Heterogeneous Networks	202141000707	Published
Chethana C	Automated Driving Assistance System	202141019449	Published

5.6 Funded Research Projects

Principal Investigator	Project Title	Funding Agency	Amount	Duration	Status
Dr. Vinay S	New Age Incubation Network (NAIN)	Government of Karnataka	₹1.2 Crore	2021–2023	Completed
Dr. Deepika	IoT Project	Government of Karnataka	₹1,30,000	2024	Ongoing
Bhavya D	Solid Waste Management System	Government of Karnataka	₹1,87,500	2024	Ongoing

Chapter 6: Professional Clubs & Chapters

6.1 IEEE PESCE Student Branch

The IEEE PESCE Student Branch, a division under IEEE Region 10, was established in 2007 and has grown to 100+ active members. It organises technical events, seminars, workshops, and social activities to enhance academic, professional, and career development.

Faculty Mentors & Contacts

Name	Role	Email	Contact
Dr. Vinay S	Branch Mentor	vinay@pesce.ac.in	+91 9986515835
Dr. Punith Kumar MB	Branch Counsellor	punithpes@gmail.com	+91 98444 82308
Dr. M C Padma	WIE Society Mentor	padmamc@pesce.ac.in	+91 9141113030
Dr. Anitha M L	COMSOC Mentor	anithaml@pesce.ac.in	+91 9945576186
Pavan Kumar K	Student Chairperson	pavankumar8656@gmail.com	+91 7603858545

Awards Won

Award	Year	Recipient
IEEE Mysore Subsection Outstanding Student Volunteer Award	2023	Aniruddh P Koundinya
IEEE Mysore Subsection Most Promising Student Branch Award	2021	PESCE Student Branch
IEEE Mysore Subsection Outstanding Student Volunteer	2021	Yeshwant L M
IEEE Mysore Subsection Outstanding Student Volunteer	2021	Skanda M K

IEEE-Funded Projects

Project Name	Developed By	Funding Body
Virobot: The Artificial Nurse Project	Manikanth Amaresh and Savadatti	IEEE India Council
Virobot Mini Prototype	Manikanth Amaresh, Savadatti, Sushma H S, Dhanush J, Preetham Kishore	IEEE Student Humanitarian Project
Internet Multi Bot	Manikanth Amaresh, Savadatti, Anup P H, Sushmitha H S, Bhoomika H J	IEEE Student Innovative Project

6.2 ISTE PESCE Chapter

PES College of Engineering became an Institutional Member of ISTE in 1986. The ISTE-PESCE Chapter was formally established in 1989. Almost all staff members are life members of ISTE. The chapter conducts short-term courses, technical workshops, and Eureka sessions annually.

ISTE President	Dr. Veena M N, Ph.D – Professor, Dept. of MCA Mob: +91 98451 42764
ISTE Secretary	Dr. Nayaka S R, Ph.D – Asst. Professor, Dept. of Mathematics Mob: +91 96116 00098
Social Links	Facebook: facebook.com/pesceiste Instagram: instagram.com/pesce.iste
Annual Reports	ISTE PESCE Chapter Reports: 2018, 2019, 2020, 2021, 2023, 2024 available on PESCE website

6.3 CSI Student Chapter

PESCE attained CSI institutional membership from 1st January 2015. The CSI-Student Chapter was inaugurated on 20th November 2015, with 108+ student members. The chapter aims to create opportunities for students to interact with industry professionals and organise seminars, workshops, and competitions.

CSI Core Committee

Name	Position
Ruksar Ahmadi	Student Chair
Pratyush Kumar	Vice Chair
Rubal Guptha	Secretary
Ashwini M J	Joint Secretary
Abhisht Aharma	Treasurer
Vidyashree H K	SAC Co-ordinator
Archana Muralidharan	Communication Lead
Yashwini S	Designer
Shreeraksha	Assistant Designer
Kshitiz Raj	Web Master
Vishal Soni	Assistant Web Master
Albert Abraham	Director of Technical Activities
Vidya	Director of Social Activities
Vyjayanthi	Membership Head

6.4 GLUG – GNU/Linux Users' Group

GLUG PESCE promotes open-source software awareness and free software values among students. Supported by the Free Software Movement of Karnataka (FSMK), it conducts workshops, installation fests, and technical sessions. GLUG is co-ordinated by Dr. Minavathi, Professor, PESCE.

Date	Activity	Resource Person
17-08-2015	Orientation	Mr. Rakesh P, FSMK / SPAN INFOTECH, Bangalore
29-08-2015	OSM Tracking Using Android App	Rakesh P and Shijil, FSMK Organization
29-08-2015	Session on PHP	Mr. Manu, VTIGER, Bengaluru (Alumni PESCE)
12-09-2015	Xubuntu, Linux Installation Fest	Rakesh, Manu, Vishal, Shijil – FSMK
03-10-2015	Introduction to Basic Linux Commands	Ramyashree P M and Vaibhav Vishal, GLUG Volunteers
17-18 Oct 2015	DRUPAL Workshop	Rameez Thonnakkal & Shijil, FSMK
19-10-2016	Introduction to Linux	Shubodip Ghosh and AmoghaVarsha, GLUG Core
06-11-2016	Java Chat Application	Nootan, GLUG Core Member
03-02-2017	Linux Installation Fest	Praveen Kumar B N and Amoghavarsha, GLUG Core
09-02-2017	Introduction to OSS and Free Software	Anup Pravin, Alumni PESCE / FSMK Member
16-02-2017	Basic Commands in Linux	Harshitha R, Anant Mohan, Sindhu – GLUG & FSMK
01-03-2017	Shell Scripting	Amoghavarsha, GLUG Core Member
16-03-2017	Shell Scripting	Amoghavarsha, GLUG Core Member
20-24 Jan 2019	Residential Camp – FSMK Winter Camp '19	Dr. Richard Stallman, GLUG & FSMK Core Members

6.5 Hack Club PESCE

Hack Club PESCE is a student-run technical club with the motto 'By the students – For the students.' Members learn by building projects in areas including IoT, Web Development, AI, Machine Learning, Robotics, and Photography. The club is an open community where passion for making is the only entry requirement.

Website	https://www.hcpes.ml
Focus Areas	IoT, Web Dev, AI, Machine Learning, Robotics, Photography, Open Projects
Community	Completely open – no membership fee; interest and passion required

Chapter 7: Centres of Excellence & Laboratories

7.1 Centre of Excellence – Internet of Things (IoT)

The Centre of Excellence on Internet of Things (IoT) was established at the Department of Computer Science and Engineering to facilitate projects and research by UG, PG, and Ph.D. students and faculty. The lab addresses real-life and research problems using IoT-based solutions.

Equipment Available

- Zolertia Gold Pack Devices (10 units) and Zolertia Gateway
- Arduino UNO R3 (ATmega328P), Arduino Yun, Raspberry Pi 2 Model B
- Sensors: Ultrasonic Range Finder, Microphone Sound Sensor, MQ-7 CO Sensor, PIR Motion Sensor, Soil Moisture Sensor, Water Level Switch, Flame Sensor, Fingerprint Sensor, GPS Module, Wind Speed/Direction Sensor, EMG Detector, GSR Sensor
- Networking: Ethernet W5100 Shield with Micro SD, Serial Camera Kit
- Electronic Components: Breadboards, Jumper Cables, LEDs, Resistors, Capacitors, Transistors
- Medical Sensors: Finger-clip Heart Rate Sensor, Adult reusable NIBP cuff
- Water Flow Sensor, Firebase Integration Components

7.2 Data Science & AI/ML Lab

A dedicated Data Science Lab is available for faculty and students of any engineering department. It supports projects in Natural Language Processing, Computer Vision, Financial Analytics, Marketing AI, Healthcare, and Transportation, covering the full ML pipeline from data collection to model deployment.

7.3 Network Forensics Laboratory

The Network Forensics Lab supports research in digital forensics, intrusion detection, and cybersecurity. It enables detection of TCP SYN flooding, UDP flooding, analysis of routing protocols, packet filtering, and round-trip-time estimation. The lab focuses on deformed packet handling, aggregation, and reassembly for real-time network analysis.

7.4 Computing Infrastructure

Total Labs	Multiple labs spread across the department (20+ systems per lab)
Network	LAN-connected, internet-enabled computing facilities
Software Tools	Python, Java, C/C++, MATLAB, R, TensorFlow, PyTorch, Hadoop, SQL databases, EDA tools
Cloud Access	Access to AWS, Azure, and Google Cloud platforms for student projects
OS Platforms	Windows, Ubuntu/Linux, ROS for Robotics projects

Chapter 8: Placements & Career Development

The Department of CSE has a strong placement record with graduates joining top-tier IT, product, and service companies. The Training and Placement Cell, headed by Dr. Vinay S (Vice Principal & Placement Officer), works closely with the department to provide placement support, mock interviews, and career guidance.

8.1 Key Recruiters

Company	Sector
Tata Consultancy Services (TCS)	IT Services
Accenture	IT Consulting & Services
Amazon	E-Commerce / Cloud / Product
Cisco Systems	Networking & Technology
Flipkart	E-Commerce / Product
Amadeus	Travel Technology
Tech Mahindra	IT Services
Infosys	IT Services
Wipro	IT Services
Morgan Stanley	Financial Services / Technology
Byju's	EdTech
30+ more companies	IT / Product / Core / Startups

8.2 Career Development Activities

- Regular mock interviews, group discussions, and aptitude training sessions.
- Resume building and LinkedIn optimisation workshops.
- Industry visits and exposure to live enterprise environments.
- STUDENTPRENEUR – Entrepreneurship initiative with IIC.
- Alumni technical talks and placement guidance sessions.
- Skill Development Programmes: Python, Full Stack, AI/ML, VLSI, and more.
- Faculty Development Programme: Microsoft Azure AI Engineer Associate.
- MongoDB Online Orientation Programme for students and faculty (Aug 2025).

Chapter 9: Contact Information

Department	Computer Science & Engineering
Head of Department	Dr. Anitha M L anithaml@pesce.ac.in +91 9945576186
Placement Officer	Dr. Vinay S vinay@pesce.ac.in +91 9986515835
Department Page	https://pesce.ac.in/department-computer-science.php
College Address	K V Shankaragowda Rd, PES College Campus, Mandya, Karnataka – 571401
College Phone	08232 – 220043
College Mobile	+91 9448282588
Admissions Email	admissions@pesce.ac.in
Website	https://pesce.ac.in
IEEE Branch	http://www.ieeePesce.com
Hack Club	https://www.hcpes.ml
NPTEL	https://nptel.ac.in
ERP Portal	https://pesgroup.dhi-edu.com